

torch.div#

<https://pytorch.org/docs/stable/generated/torch.div.html>

Description:

Divides each element of the input `input` by the corresponding element of `other`. By default, this performs a “true” division like Python 3. See the `rounding_mode` argument for floor division. Supports broadcasting to a common shape, type promotion, and integer, float, and complex inputs. Always promotes integer types to the default scalar type. `input` (Tensor) – the dividend

Function Signature

```
torch.div(input, other, *, rounding_mode=None, out=None) → Tensor#
```

Parameters

Keyword Arguments

Parameters

Keyword

`rounding_mode` (str, optional) – Type of rounding applied to the result: `None` - default behavior. Performs no rounding and, if both input and other are integer types, promotes the inputs to the default scalar type. Equivalent to true division in Python (the `/` operator) and NumPy's `np.true_divide`. `"trunc"` - rounds the results of the division towards zero. Equivalent to C-style integer division. `"floor"` - rounds the results of the division down. Equivalent to floor division in Python (the `//` operator) and NumPy's `np.floor_divide`. `out` (Tensor, optional) – the output tensor.

Code Examples

```
>>> x = torch.tensor([ 0.3810,  1.2774, -0.2972, -0.3719,
0.4637])
>>> torch.div(x, 0.5)
tensor([ 0.7620,  2.5548, -0.5944, -0.7438,  0.9274])

>>> a = torch.tensor([[ -0.3711, -1.9353, -0.4605, -0.2917],
...                    [ 0.1815, -1.0111,  0.9805, -1.5923],
...                    [ 0.1062,  1.4581,  0.7759, -1.2344],
...                    [ -0.1830, -0.0313,  1.1908, -1.4757]])
>>> b = torch.tensor([ 0.8032,  0.2930, -0.8113, -0.2308])
>>> torch.div(a, b)
tensor([[ -0.4620, -6.6051,  0.5676,  1.2639],
        [ 0.2260, -3.4509, -1.2086,  6.8990],
        [ 0.1322,  4.9764, -0.9564,  5.3484],
        [ -0.2278, -0.1068, -1.4678,  6.3938]])

>>> torch.div(a, b, rounding_mode='trunc')
tensor([[ -0., -6.,  0.,  1.],
        [ 0., -3., -1.,  6.],
        [ 0.,  4., -0.,  5.],
        [ -0., -0., -1.,  6.]])

>>> torch.div(a, b, rounding_mode='floor')
tensor([[ -1., -7.,  0.,  1.],
        [ 0., -4., -2.,  6.],
        [ 0.,  4., -1.,  5.],
        [ -1., -1., -2.,  6.]])
```

Notes & Warnings

NOTE

Note By default, this performs a “true” division like Python 3. See the `rounding_mode` argument for floor division.

Detailed Documentation

Documentation

Divides each element of the input input by the corresponding element of other.

By default, this performs a “true” division like Python 3. See the `rounding_mode` argument for floor division.

Supports broadcasting to a common shape, type promotion, and integer, float, and complex inputs. Always promotes integer types to the default scalar type.

`input` (Tensor) – the dividend

`other` (Tensor or Number) – the divisor

`rounding_mode` (str, optional) –

Type of rounding applied to the result:

None - default behavior. Performs no rounding and, if both `input` and `other` are integer types, promotes the inputs to the default scalar type. Equivalent to true division in Python (the `/` operator) and NumPy's `np.true_divide`.

"trunc" - rounds the results of the division towards zero. Equivalent to C-style integer division.

"floor" - rounds the results of the division down. Equivalent to floor division in Python (the `//` operator) and NumPy's `np.floor_divide`.

`out` (Tensor, optional) – the output tensor.

